Service failures of learned multi-agent signal controllers on a two-intersection corridor: an evaluation with optimised fixed plans and paired protection ablations

[To be completed: authors and affiliations]

Draft of October 31, 2026 — development results; not peer reviewed

Abstract

We study three value-based multi-agent reinforcement learning controllers (QMIX, VDN and independent DQN) on a SUMO corridor of two signalised intersections spaced 300 m apart, with decisions every 5 s between extending the current green and switching through a 3 s yellow. The evaluation keeps every run, including runs that fail to clear the network, and compares the learned policies with fixed plans optimised on separate traffic samples, one of them re-optimised under the same 7 s minimum green. On five development traffics, the historical fixed plan obtains a lower mean waiting burden than every learned method, and the re-optimised constrained plan remains competitive with the historical QMIX policies. Inspecting the failures shows a mechanism that mean metrics hide: after the demand ends, a greedy policy can enter a nearly stationary state in which an isolated vehicle on a red approach is never served. We then add two protections at inference time, to policies trained without them — a 7 s minimum green and a service guarantee whose counter is provably bounded by 127 s — and measure their effects on the same models and traffics (600 paired evaluations). The guarantee cleared all six failed runs, intervening once per run and only after the demand had ended, so its cost during the demand was not exercised; the minimum green raised the waiting burden of every QMIX and VDN policy (by 1.2–2.3 s per vehicle) and had mixed effects on IDQN. The comparison under this common constraint is asymmetric: the fixed plan was re-optimised for it, whereas the historical policies were trained without it and constrained only at evaluation, without retraining; none of their 299 runs that cleared matched that plan (the 300th failed to clear). Single-seed pilots trained with both protections give an exploratory, unfavourable signal; they do not indicate how a full multi-seed training campaign would turn out. The results are development evidence on already-inspected traffics; a robustness study on new traffics is prepared, and a decision on retraining is deferred.

1 Introduction

Deep reinforcement learning (RL) controllers for traffic signals are commonly reported to reduce delays relative to fixed-time or actuated control [6, 7, 9, 17, 26–28]. Two features of how these comparisons are usually made limit what they show. First, the reference plans are often not optimised under the same operational constraints as the learned controller, which has been pointed out for fixed-time baselines in particular [3, 4]. Second, the headline metrics are averages over completed trips or over successful episodes; a controller that occasionally leaves a vehicle waiting indefinitely can look good on such averages, and evaluation practice in deep RL more broadly is sensitive to such reporting choices [1, 10, 12, 25].

We examine these two issues on a deliberately small and fully controlled case: two intersections, three value-based multi-agent methods with centralised (QMIX [20], VDN [21]) or independent (IDQN [16, 22]) training, and fixed plans searched on separate traffic samples. The questions are:

1. On the same traffics, how do the learned controllers compare with optimised fixed plans, including a plan optimised under the same minimum green?
2. When a learned controller fails to clear the network, what happens, and is it visible in the usual metrics?
3. What do two simple protections — a minimum green and a service guarantee — change when added at evaluation time to policies trained without them, on efficiency and on service?

The contribution is empirical and methodological. Minimum greens and action masks are standard [2, 11]; we do not claim novelty for them. What we report is (i) a hash-verified comparison with constrained optimised fixed plans, (ii) a documented failure mechanism of greedy learned policies after the demand ends, and (iii) a paired measurement, on identical models and traffics, of what each protection costs during the demand and what it changes afterwards.

2 Related work

Learned signal control. Single-agent and multi-agent deep RL controllers include IntelliLight [26], PressLight [27], CoLight [28], FRAP [30], MPLight [6] and multi-agent actor–critic methods [7]; earlier multi-agent work includes MARLIN-ATSC [8]. Surveys [9, 17] and benchmark libraries [3, 5, 15] document the field. PyTSC [5] compares IQL, VDN and QMIX on SUMO grids; in generic cooperative tasks, independent learning is often competitive with value factorisation [18]. [To be completed: confirm the network, constraint and baseline details of these works from the full texts (only abstracts and metadata were checked for this draft)].

Model-based adaptive control and service. Max pressure has throughput guarantees [24]; its cyclical variant prevents a phase from being skipped indefinitely [13]. Queue-based pressure control can leave a lightly loaded movement waiting, which motivated delay-based back-pressure with fairness properties [29]. Fairness has also been added to the RL objective [19].

Constraints on learned actions. Masking invalid actions and shielding a learner with an external monitor are established [2, 11]. Our masks are simple instances of this idea; their interest here lies in the paired measurement, not in the mechanism.

3 Environment and controllers

Network and demand. The SUMO [14] network has two signalised intersections (J1, J2) on a two-lane main axis; the central link is 279.2 m long (300 m between intersections). Each intersection has two phases, main (H) and secondary (V). Demand is generated over one hour by a conditioned multinomial model with fixed counts: 900 (west→east) and 700 (east→west) main-axis vehicles and 300 per secondary stream (four streams), 2,800 vehicles in total. Simulation continues until the network clears or until 7,200 s.

Learned controllers. Each agent observes eight normalised local features (halting queues on H and V, current phase, elapsed green, recent arrival rates on H and V, downstream occupancies). Actions are EXTEND (5 s more green) and SWITCH (3 s yellow then 2 s of the other green), decided synchronously every 5 s. The shared reward is the negative number of stopped vehicle-seconds plus due-but-not-inserted vehicle-seconds over the 5 s block. Each agent uses a multilayer perceptron (8–128–128–2); QMIX adds a monotonic mixer conditioned on the joint 16-dimensional observation (mixing width 32, hypernetwork width 64) [20]; VDN sums the agent values [21]; IDQN trains each agent independently. All use double-Q targets [23], a 200,000-transition replay, batch 256, learning rate $5 \cdot 10^{-4}$, $\gamma = 0.99$ per 5 s, ε decaying from 1 to 0.05 over 250,000 transitions, and 360,000 transitions per training. Ten training seeds per method were run; one checkpoint per training was selected by a fixed rule on separate validation traffics (eligibility on hard validity gates, then minimum mean waiting burden, then minimum time loss, then earliest checkpoint).

Fixed plans. Two-phase fixed plans with common cycle, green split and offset were searched on separate design traffics and selected on separate validation traffics. The historical plan has a 19 s cycle with 8 s and 5 s greens and a 9 s offset. Because a 5 s green violates the 7 s minimum adopted below, the search was repeated with greens of at least 7 s, giving a 25 s cycle with 12 s and 7 s greens and zero offset (selection on the same design and validation samples as the original search, so this selection is not an independent validation).

Protections. *Minimum green* $G = 7$ s forbids SWITCH while the current green has lasted less than G; with the 5 s clock and the 3+2 s switch, realised greens are $2 + 5k$ s, and 7 s is the smallest realisable value above 5 s. *Service guarantee*: for each red direction a counter increases every second while at least one vehicle halts on its observed incoming lanes, and resets on green or when the queue empties; when it reaches $T = 120$ s, EXTEND is forbidden. Both are applied as a mask outside the agents: the executed action is the greedy action if allowed, otherwise the other action. Under the 5 s/3+2 s executor the two masks never forbid both actions when $T \geq G$, and the counter never exceeds $T + 4 + 3 = 127$ s. The bound concerns the counter (continuous presence of a queue on a red approach), not the cumulative waiting time of a vehicle or its arrival.

4 Experimental method

Traffics. All results below use five development traffics (seeds 9301–9305), each with its own SUMO seed. They had been inspected before this study; the results are therefore development evidence and not a final test.

Indicators. J is the mean over the 2,800 scheduled vehicles of SUMO waiting time plus departure delay, defined only when the network clears; failed runs are kept and counted, never imputed. Because J is undefined for a failure, we also report the waiting burden *during the demand*,

$$D = \sum_{t=1}^{3600} \left(n_{\text{stopped}}(t) + n_{\text{pending}}(t)\right) \text{ vehicle-seconds}, \qquad \bar{D} = D/N,$$

with $N = 2{,}800$, which is defined for every run. Service is described separately by (a) the service counter, (b) the access time to the first green of a waiting direction, (c) stop-line crossings (main axis only), and (d) arrivals (per-vehicle waiting time and time loss).

Table 1: Development traffics 9301–9305. J is averaged over cleared runs only; n is the number of cleared runs. Time loss is the mean over completed vehicles of each run, averaged over the cleared runs. Pilot rows: one training seed each, trained with both protections, two checkpoints.

Controller	cleared / runs	mean J (s)	mean time loss (s)
Historical fixed plan (C 19 s, 8/5 s, offset 9 s)	5/5	2.772	11.74
Fixed plan, greens $\geq$ 7 s (C 25 s, 12/7 s)	5/5	3.156	10.13
QMIX, 10 policies, no protection	49/50	3.106	12.15
VDN, 10 policies, no protection	50/50	3.133	12.08
IDQN, 10 policies, no protection	46/50	6.123	19.54
QMIX pilot, 250k / 360k transitions	5/5, 5/5	4.698 / 4.435	14.02 / 13.57
VDN pilot, 250k / 360k	5/5, 5/5	4.324 / 4.815	13.23 / 14.59
IDQN pilot, 250k / 360k	5/5, 5/5	5.478 / 4.898	14.95 / 13.91

Ablation design. The 30 selected checkpoints (3 methods $\times$ 10 seeds) are evaluated greedily on the five traffics under four arms: no protection, guarantee only, minimum green only, both. This gives 600 evaluations in 150 blocks of four paired runs. Fifty QMIX runs without protection were taken from an earlier evaluation with the same checkpoints, traffics and simulator seeds; their files were hash-verified unchanged. An earlier technical check, whose log is not part of the present data package, found that the evaluator used for the new runs reproduces the frozen output files byte for byte in the absence of protection. [To be completed: attach the log of that check to the replication package] Effects are paired differences within blocks; ΔJ uses only blocks where both runs clear, and their number is reported. The analysis is descriptive: models and traffics are crossed factors and no pooled hypothesis test is made.

5 Results

5.1 Learned controllers and fixed plans on the same traffics

Table 1 and Figure 1 report every evaluation on the five traffics. The historical fixed plan has the lowest mean J (2.772 s). Re-optimising the fixed plan under the 7 s minimum green raises its J to 3.156 s while lowering its mean time loss from 11.74 to 10.13 s, an example of the two indicators disagreeing. The ten historical QMIX policies obtain 3.106 s on 49 valid runs out of 50; VDN 3.133 s (50/50) and IDQN 6.123 s (46/50). No learned run that cleared has a lower J than the historical fixed plan on the same traffic (145 of the 150 unprotected runs cleared; over the four arms of Section 5.3, 594 of 600). Without the minimum green, QMIX and VDN obtain a lower J than the constrained plan in 33 of 49 and 30 of 50 runs, but they do so with 2 s greens (57–61 % of their greens during the demand) that the constrained plan is not allowed; their mean time loss is never lower than that plan's. Single-seed pilots trained with both protections obtained mean values of 4.32–5.48 s, above both fixed plans. This is an exploratory, unfavourable signal only: one training seed per method, no pairing by seed with the ten historical policies, and no indication of how a full multi-seed campaign would turn out.

5.2 Anatomy of a clearance failure

The QMIX failure (one policy, one traffic) leaves a single vehicle: the last vehicle of a secondary stream, inserted at 3,578 s, halted at the stop line from 3,605 s to the 7,200 s cut-off, with no other

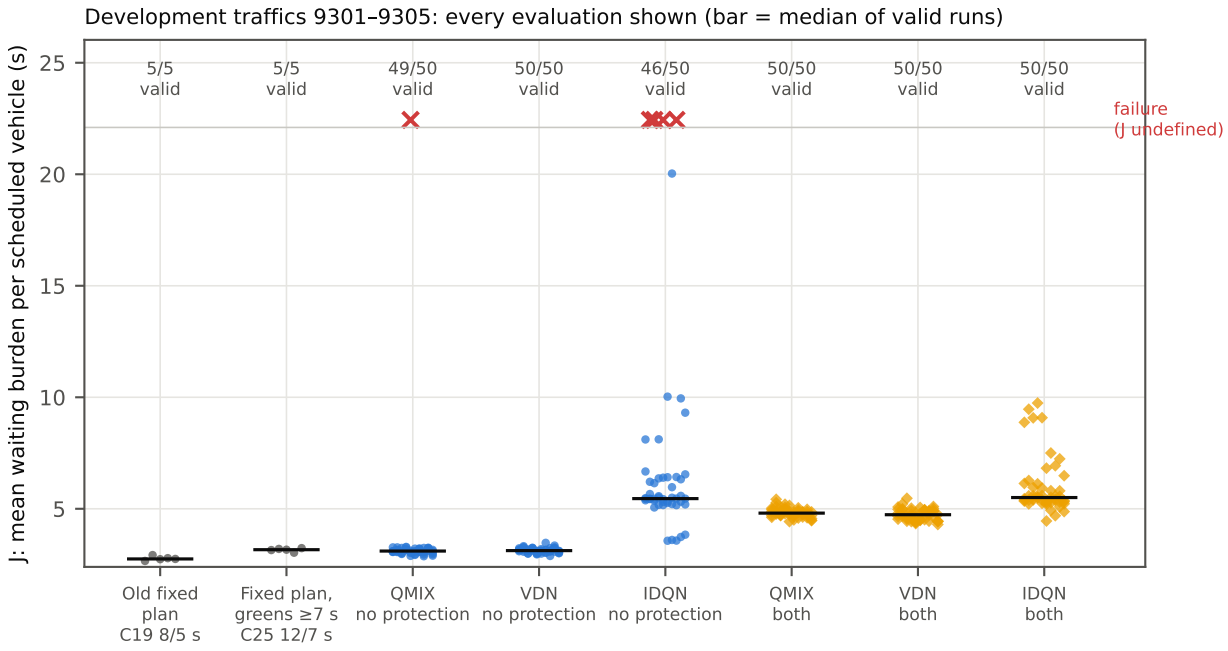


Figure 1: Every evaluation on the five development traffics. Failed runs are drawn as crosses in a separate band and are not averaged; counts above each group.

vehicle in the network after 3,679 s and no insertion backlog. A 2 s secondary green ended just before the vehicle reached the stop line; afterwards the intersection's observation became nearly stationary (one halted vehicle on the secondary approach, no arrivals, main green), and only the elapsed-green feature changed. Along the logged trajectory the greedy margin $Q(\text{EXTEND}) - Q(\text{SWITCH})$ grew from 0.24 to 0.62 as that feature saturated, so the policy extended the main green at all 718 remaining decisions. The same policy cleared another traffic only after a 338 s wait at the other intersection, where the same feature drifted the margin across zero. Every failure in this study (one QMIX and four IDQN runs without protection, one IDQN run under the minimum green alone) concerns the last one or two vehicles of a secondary stream, left after the demand: one unfinished vehicle in five runs, two in one run. The executed phases were logged as commanded by the controller; a read-back of SUMO's signal state is available only for the new evaluations, where it showed no mismatch.

5.3 Effects of the protections

Guarantee. With the guarantee alone, all 150 runs cleared (QMIX 50/50, VDN 50/50, IDQN 50/50, against 49, 50 and 46 without protection). It intervened once in each of 13 runs, always after 3,600 s; it never intervened during the demand. Consequently the waiting burden during the demand was identical to the unprotected run in all 150 pairs ($\Delta D = 0$), and ΔJ was zero except in pairs where a late intervention shortened a wait (at most -0.075 s). The maximum counter was 127 s, as the bound requires, but the largest cumulative waiting time of a vehicle reached 169 s: a vehicle can wait at both intersections, and the counter resets when a queue empties, so the bound does not apply to vehicle waits.

Minimum green. For QMIX and VDN the effect was regular and large: $\Delta D > 0$ in 100/100 pairs, ΔJ between +1.29 and +2.28 s (QMIX, 49 comparable pairs) and between +1.20 and +2.30 s (VDN, 50 pairs), and higher time loss in every pair. The mask overrode about 350 decisions per run (QMIX median). For IDQN the effect was variable: ΔD was positive in 25 and negative in 25 of 50 pairs (median +56 veh s, range −35,004 to +4,514), the median ΔJ was −0.06 s over 45 comparable pairs, and the minimum green cleared four failed runs but created one.

Both protections. The combination cleared 150/150 runs. Because the guarantee never acted during the demand, the interaction contrast on D was zero in all 150 blocks, and on J it was zero or below 0.14 s in absolute value.

Re-optimised fixed plan versus historical policies constrained at evaluation. The two sides of this comparison were not treated alike: the fixed plan was re-optimised for the 7 s minimum green, whereas the learned policies are the historical ones, trained without it and constrained only at evaluation, without retraining. Under the 7 s minimum green (arms with the minimum green), 299 of these 300 runs cleared, and none of these 299 had a lower J or a lower mean time loss than the constrained fixed plan on the same traffic (mean differences +1.58 to +2.82 s in J by method). The remaining run (IDQN, minimum green only) failed to clear; it has no J and is not counted as a comparison.

6 Discussion

Failures are a property of the evaluation, not noise. The failures are rare (6 of 300 runs without the guarantee) but systematic in kind: a lone vehicle after the demand, facing a greedy policy whose inputs barely change. Metrics computed on completed vehicles or on cleared runs do not register them; the failed QMIX run has a mean time loss of 11.9 s for its completed vehicles while one vehicle waits 3,600 s. Reporting the number of failures, keeping them out of averages, and using an indicator defined for every run (D) change what can be concluded.

The guarantee is cheap here because it is rarely needed. On these traffics the guarantee repaired every failure at no measurable cost, but only because it acted after the demand. Whether it is cheap when secondary demand is light but persistent during a loaded period is untested; the prepared robustness study targets exactly that.

The minimum green exposes a dependence on very short greens. QMIX and VDN policies spend most of their greens in the 2 s minimum the executor allows. Imposing 7 s at inference removes that behaviour and costs 1.2–2.3 s of waiting per vehicle. This measures the brittleness of these policies to a constraint they were not trained with, not the value of the constraint itself, nor what a policy trained under it can achieve. The single-seed pilots trained with the constraint are an exploratory, unfavourable signal on that question; they do not predict whether a full campaign would close the gap to the re-optimised fixed plan or not.

Relation to the literature. The comparison supports the concern that learned controllers are often compared with weak references [4]: here, a fixed plan re-optimised under the same constraint is not beaten by any cleared run of the historical policies constrained at evaluation (299 of 300 runs; the other failed to clear). Whether policies trained under the constraint would do better is not

established here. The lone-vehicle failure is a learned-controller counterpart of the starvation that queue-based pressure control can produce for lightly loaded movements [29].

7 Limitations and scope

- One corridor geometry and one demand level; five development traffics that had already been inspected.
- The protections were added at evaluation time to policies trained without them; the effect of training with them, and any effect of cooperation, are not measured here.
- The constrained fixed plan was selected on the same design and validation samples as the original plan.
- The comparison under the common constraint is asymmetric: a fixed plan re-optimised for the constraint against learned policies constrained only at evaluation. Policies trained under the constraint exist only as single-seed pilots.
- The 127 s bound applies to the service counter under the stated executor assumptions; it is not a bound on a vehicle's total waiting or travel time.
- Stop-line crossings are logged for the main axis only.

8 Reproducibility

Code, plans fixed before the runs, run ledgers with checkpoint and manifest hashes, derived data and figure scripts are provided (Appendix A). Software: Python 3.7.16, NumPy 1.21.6, PyTorch 1.13.1 (CPU), SUMO 1.25.0, Windows.

Prepared but not executed

A robustness validation on new traffics designed to stress service (persistent sparse secondary demand, time-varying demand, isolated vehicles at the end of the demand), and a retraining campaign with the protections, with new validation and final-test traffics, are specified with their analysis rules and costs. The decision to run the retraining campaign will be taken after the robustness study and a full reading of the closest works; [To be completed: results].

A Technical traceability

- Frozen learning and evaluation code: commit `8c14f32`; protected evaluator (masks, counters, signal read-back): commit `3611383`, whose frozen part is identical to `8c14f32`.
- Ablation plan fixed before the runs: fingerprint `c9a4c6d8...` (600 cells; 550 new, 50 reused with file hashes recorded and re-verified after the runs); analysis and indicator code committed before the runs (study workspace commit `e0557af`).
- Checkpoints: the 30 selected checkpoints verified by SHA-256 against the selection records before copying; traffic manifests verified identical across the two earlier studies that used them.

- Checks after the runs: 0 SUMO signal mismatches in 550 new runs; counters reconstructed from logs equal to the logged counters in 100/100 runs; counter ≤ 127 s in every run with the guarantee; realised greens ≥ 7 s in every run with the minimum green.
- Figures are produced by `figures/make_figures.py`; the plotted values are in `figures/data/`.
- Tests: the 31 tests of the constrained fixed-plan search were run on the Windows stack above (terminal output "Ran 31 tests in 1548.888s", "OK", exit status 0, transmitted by the operator; no log file of that run is retained); the indicator and robustness tools have 11 unit tests run on the Windows stack (log retained) and 10 tests of the robustness analysis on synthetic outputs.

References

[1] Rishabh Agarwal et al. Deep reinforcement learning at the edge of the statistical precipice. In *Advances in Neural Information Processing Systems*, volume 34, 2021.

[2] Mohammed Alshiekh, Roderick Bloem, Rüdiger Ehlers, Bettina Könighofer, Scott Niekum, and Ufuk Topcu. Safe reinforcement learning via shielding. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 32, 2018. doi: 10.1609/aaai.v32i1.11797.

[3] James Ault and Guni Sharon. Reinforcement learning benchmarks for traffic signal control. In *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*, 2021.

[4] James Ault, Josiah P. Hanna, and Guni Sharon. Learning an interpretable traffic signal control policy. In *Proceedings of the 19th International Conference on Autonomous Agents and MultiAgent Systems (AAMAS)*, 2020. arXiv:1912.11023.

[5] Rohit Bokade and Xiaoning Jin. PyTSC: A unified platform for multi-agent reinforcement learning in traffic signal control. *Sensors*, 25(5):1302, 2025. doi: 10.3390/s25051302.

[6] Chacha Chen, Hua Wei, Nan Xu, Guanjie Zheng, Ming Yang, Yuanhao Xiong, Kai Xu, and Zhenhui Li. Toward a thousand lights: Decentralized deep reinforcement learning for large-scale traffic signal control. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 34, 2020. doi: 10.1609/aaai.v34i04.5744.

[7] Tianshu Chu, Jie Wang, Lara Codeca, and Zhaojian Li. Multi-agent deep reinforcement learning for large-scale traffic signal control. *IEEE Transactions on Intelligent Transportation Systems*, 21(3):1086–1095, 2020. doi: 10.1109/TITS.2019.2901791.

[8] Samah El-Tantawy, Baher Abdulhai, and Hossam Abdelgawad. Multiagent reinforcement learning for integrated network of adaptive traffic signal controllers (MARLIN-ATSC). *IEEE Transactions on Intelligent Transportation Systems*, 14(3), 2013. doi: 10.1109/TITS.2013.2255286.

[9] Ammar Haydari and Yasin Yilmaz. Deep reinforcement learning for intelligent transportation systems: A survey. *IEEE Transactions on Intelligent Transportation Systems*, 23(1), 2022. doi: 10.1109/TITS.2020.3008612.

[10] Peter Henderson, Riashat Islam, Philip Bachman, Joelle Pineau, Doina Precup, and David Meger. Deep reinforcement learning that matters. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 32, 2018. doi: 10.1609/aaai.v32i1.11694.

[11] Shengyi Huang and Santiago Ontañón. A closer look at invalid action masking in policy gradient algorithms. In *The International FLAIRS Conference Proceedings*, volume 35, 2022. doi: 10.32473/flairs.v35i.130584.

[12] Jayawardana et al. The impact of task underspecification in evaluating deep reinforcement learning. In *Advances in Neural Information Processing Systems*, 2022. arXiv:2210.08607.

[13] Michael W. Levin, Jeffrey Hu, and Michael Odell. Max-pressure signal control with cyclical phase structure. *Transportation Research Part C: Emerging Technologies*, 2020. doi: 10.1016/j.trc.2020.102828.

[14] Pablo Alvarez Lopez, Evamarie Wiessner, Michael Behrisch, Laura Bieker-Walz, Jakob Erdmann, Yun-Pang Flotterod, Robert Hilbrich, Leonhard Lucken, Johannes Rummel, and Peter Wagner. Microscopic traffic simulation using SUMO. In *21st International Conference on Intelligent Transportation Systems (ITSC)*, 2018. doi: 10.1109/ITSC.2018.8569938.

[15] Hao Mei, Xiaoliang Lei, Longchao Da, Bin Shi, and Hua Wei. LibSignal: an open library for traffic signal control. *Machine Learning*, 2024. doi: 10.1007/s10994-023-06412-y.

[16] Volodymyr Mnih, Koray Kavukcuoglu, David Silver, Andrei A. Rusu, Joel Veness, Marc G. Bellemare, Alex Graves, Martin Riedmiller, Andreas K. Fidjeland, Georg Ostrovski, Stig Petersen, Charles Beattie, Amir Sadik, Ioannis Antonoglou, Helen King, Dharshan Kumaran, Daan Wierstra, Shane Legg, and Demis Hassabis. Human-level control through deep reinforcement learning. *Nature*, 518:529–533, 2015. doi: 10.1038/nature14236.

[17] Mohammad Noaeen, Atharva Naik, Liana Goodman, Jared Crebo, Taimoor Abrar, Zahra Shakeri Hossein Abad, Ana L.C. Bazzan, and Behrouz Far. Reinforcement learning in urban network traffic signal control: A systematic literature review. *Expert Systems with Applications*, 2022. doi: 10.1016/j.eswa.2022.116830.

[18] Georgios Papoudakis, Filippos Christianos, Lukas Schäfer, and Stefano V. Albrecht. Benchmarking multi-agent deep reinforcement learning algorithms in cooperative tasks. In *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*, 2021. arXiv:2006.07869.

[19] Majid Raeis and Alberto Leon-Garcia. A deep reinforcement learning approach for fair traffic signal control. In *IEEE International Intelligent Transportation Systems Conference (ITSC)*, 2021. doi: 10.1109/ITSC48978.2021.9564847.

[20] Tabish Rashid, Mikayel Samvelyan, Christian Schroeder de Witt, Gregory Farquhar, Jakob Foerster, and Shimon Whiteson. Monotonic value function factorisation for deep multi-agent reinforcement learning. *Journal of Machine Learning Research*, 21(178):1–51, 2020.

[21] Peter Sunehag et al. Value-decomposition networks for cooperative multi-agent learning based on team reward. In *Proceedings of the 17th International Conference on Autonomous Agents and MultiAgent Systems (AAMAS)*, 2018. arXiv:1706.05296.

[22] Ardi Tampuu, Tambet Matiisen, Dorian Kodelja, Ilya Kuzovkin, Kristjan Korjus, Juhan Aru, Jaan Aru, and Raul Vicente. Multiagent cooperation and competition with deep reinforcement learning. *PLOS ONE*, 12(4):e0172395, 2017. doi: 10.1371/journal.pone.0172395.

[23] Hado Van Hasselt, Arthur Guez, and David Silver. Deep reinforcement learning with double Q-learning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 30, 2016. doi: 10.1609/aaai.v30i1.10295.

[24] Pravin Varaiya. Max pressure control of a network of signalized intersections. *Transportation Research Part C: Emerging Technologies*, 36:177–195, 2013. doi: 10.1016/j.trc.2013.08.014.

[25] Varela et al. A methodology for the development of RL-based adaptive traffic signal controllers, 2021. arXiv:2101.09614; final publication venue to be verified.

[26] Hua Wei, Guanjie Zheng, Huaxiu Yao, and Zhenhui Li. IntelliLight: A reinforcement learning approach for intelligent traffic light control. In *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, 2018. doi: 10.1145/3219819.3220096.

[27] Hua Wei, Chacha Chen, Guanjie Zheng, Kan Wu, Vikash Gayah, Kai Xu, and Zhenhui Li. PressLight: Learning max pressure control to coordinate traffic signals in arterial network. In *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, 2019. doi: 10.1145/3292500.3330949.

[28] Hua Wei, Nan Xu, Huichu Zhang, Guanjie Zheng, Xinshi Zang, Chacha Chen, Weinan Zhang, Yanmin Zhu, Kai Xu, and Zhenhui Li. CoLight: Learning network-level cooperation for traffic signal control. In *Proceedings of the 28th ACM International Conference on Information and Knowledge Management*, 2019. doi: 10.1145/3357384.3357902.

[29] Jian Wu, Dipak Ghosal, Michael Zhang, and Chen-Nee Chuah. Delay-based traffic signal control for throughput optimality and fairness at an isolated intersection. *IEEE Transactions on Vehicular Technology*, 67(2):896–909, 2018. doi: 10.1109/TVT.2017.2760820.

[30] Guanjie Zheng, Yuanhao Xiong, Xinshi Zang, Jie Feng, Hua Wei, Huichu Zhang, Yong Li, Kai Xu, and Zhenhui Li. Learning phase competition for traffic signal control. In *Proceedings of the 28th ACM International Conference on Information and Knowledge Management*, 2019. doi: 10.1145/3357384.3357900.

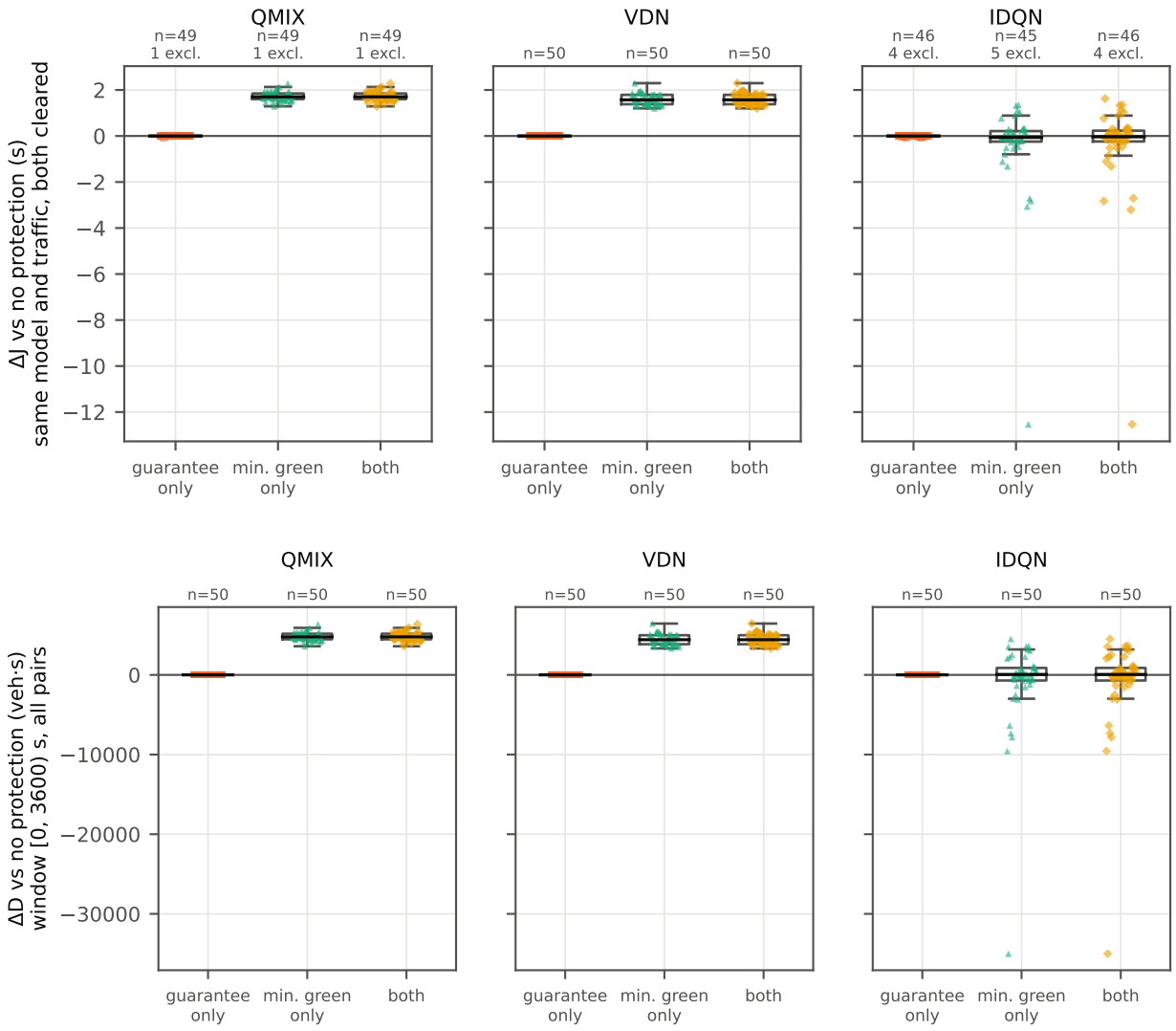


Figure 2: Paired effects of each protection relative to the same model and traffic without protection. Top: ΔJ (pairs where both runs clear; excluded pairs counted). Bottom: ΔD over the demand window (all pairs).

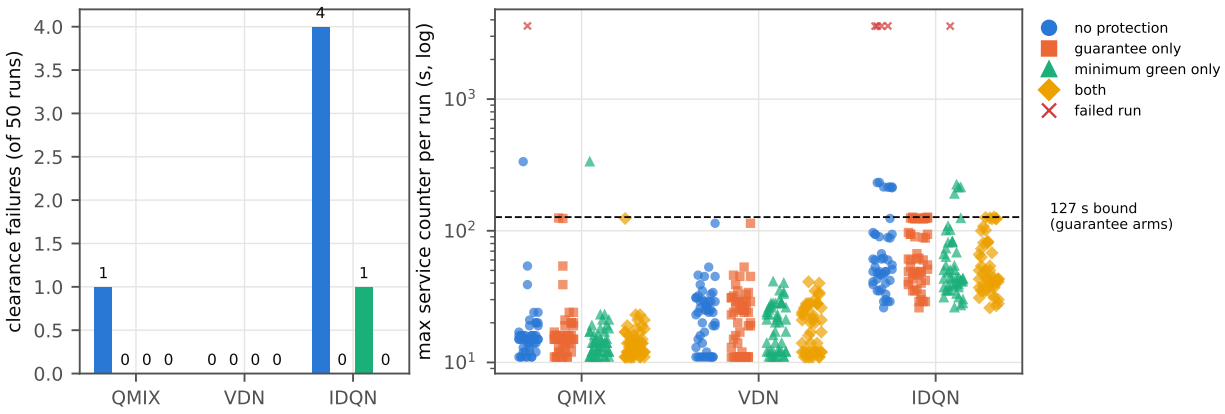


Figure 3: Left: clearance failures per method and arm. Right: maximum service counter per run (log scale); the dashed line is the 127 s bound that holds for the arms with the guarantee.

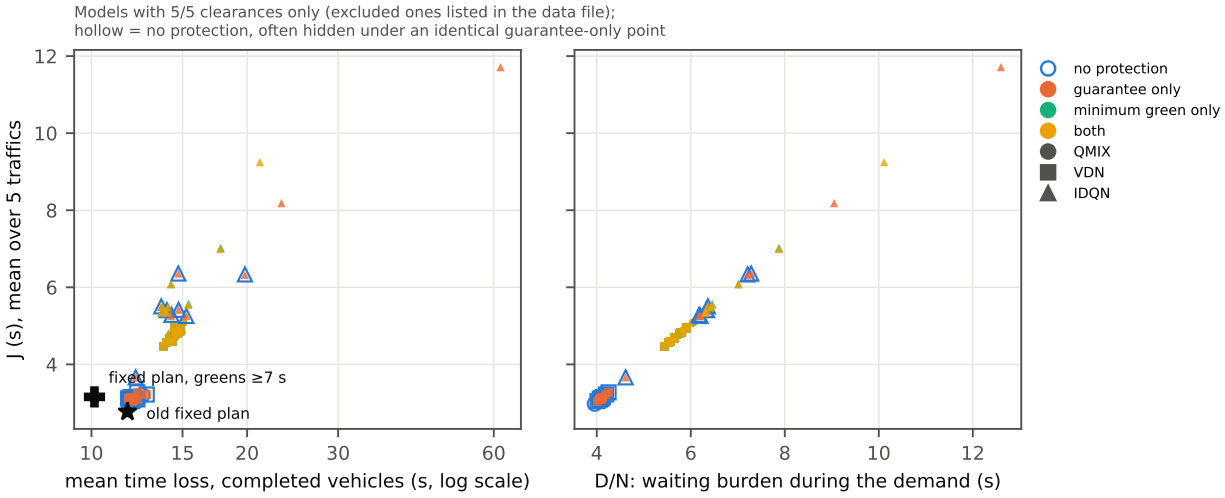


Figure 4: Indicator trade-offs: mean J against mean time loss of completed vehicles (left) and against $\bar{D}$ (right), for models that cleared all five traffics; excluded models are listed in the data file.